SAI HARSHITH REDDY SANKALAMADDI

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PROFESSIONAL SUMMARY

Analytical Computer Science (Data Science) undergraduate with a strong dual foundation in **Software Development** and **Data Analytics**. Competent in software engineering principles, including architecting modular applications with Object-Oriented Programming (OOP) and REST APIs, alongside building predictive machine learning models and end-to-end data processing pipelines. Proven capability in translating complex technical requirements and packet-level network diagnostics into high-impact software solutions and data-driven business strategies.

TECHNICAL SKILLS

Languages: Python (Highly Proficient), SQL (Advanced), Java, C, JavaScript, HTML5/CSS3
Data Science & ML: EDA, Predictive Modeling, Feature Engineering, Pandas, NumPy, Scikit-Learn, TensorFlow, Matplotlib
Software & Web: OOP, Data Structures & Algorithms (DSA), React, REST APIs, JSON Parsing, Relational Databases (DBMS)
Tools & Methods: Git/GitHub, MS Excel, Scapy, PyShark, Apache Kafka, Apache Spark, SDLC, Agile/Scrum, Distributed ML Topologies (Federated, Ring All-Reduce, Split Learning)

EDUCATION

Avanthi Institute of Engineering & Technology — Hyderabad, India
B.Tech in Computer Science (Data Science) — Expected Graduation: May 2026
Cumulative CGPA: 8.1/10.0
Core Coursework: DBMS, Operating Systems, Computer Networks, Data Structures & Algorithms (DSA), OOP, Software Engineering.

ACADEMIC CAPSTONE PROJECT

Machine Learning in Computer Networks: Distributed Frameworks **Python, Scikit-Learn, TensorFlow, Scapy, PyShark**

- Designed a multi-modular simulation framework analyzing how ML optimizes network operations and distributed ML training topologies (Federated Learning, Ring All-Reduce, Split Learning).
- Engineered a real-time data pipeline using `scapy` and `pyshark` to capture raw packet streams, preprocess anomalies, and extract flow features (latency, throughput, packet loss).
- Deployed classification models to automate traffic categorization and flag proactive threats, replacing traditional reactive signature-based defense systems.

PERSONAL & CASE STUDY PROJECTS

Nutrition Data Calculation & Analysis System — **Python, REST APIs, OOP, Data Engineering**

- Engineered a modular application to track and calculate daily nutritional intake using high-precision OOP computational logic, reducing data retrieval latency by 20%.
- Integrated third-party REST APIs to sync real-time food datasets, parsing complex JSON payloads into optimized local relational databases.

Customer Behavior Prediction Model — **Python, Scikit-Learn, Pandas, Matplotlib**

- Developed an end-to-end predictive classification pipeline utilizing Logistic Regression to forecast customer engagement trends.
- Handled end-to-end preprocessing (imputation, scaling, encoding) on demographic datasets and visualized key data correlations using Matplotlib to support business decision-making.

Retail Business Feasibility Study — **Business Analysis, Financial Modeling, MS Excel**

- Conducted a market feasibility analysis for a retail venture, analyzing demographic density and competitor pricing to design strategic brand positioning guidelines.
- Developed multi-scenario financial forecast models in Excel to project exact break-even points and operational margins.

CERTIFICATIONS

- **Data Camp:** Python for Data Science • SQL Fundamentals • Excel Fundamentals